REVIEW

Biofuels and Electrofuels as Alternative Green Fuels for Marine Applications: A Review

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ABSTRACT

The International Maritime Organization (IMO) has imposed strict regulations to limit marine emissions because the maritime sector is expanding around the world, producing large amounts of emissions that are harmful to the atmosphere. Green alternative fuels, such as biofuels derived from biomass and electrofuels derived from syngas sources, play critical roles in meeting IMO requirements for clean energy with zero emissions. This study presents a brief review of two types of green fuels: 1) the production of biofuels from biomass sources by using various methods, such as the gasification process and the pyrolysis process, as well as the effectiveness of adding a variety of catalysts, and 2) electrofuels as a new method to oppose global warming by employing various carbon capture strategies and renewable hydrogen production based on water electrolysis. Following that, the significant effect of using these green fuels in marine applications is discussed. Overall, the primary goal of this article is to provide data for researchers and industrialists interested in biofuels and electrofuels as promising alternatives to fossil fuels. A large portion of the existing literature published in highly regarded journals, including the most recently published reports, is analyzed.

Keywords: biomass, carbon capture, hydrogen production, electrofuels

Introduction

Currently, about 27,000 aircraft and 90,000 ships, in addition to 1.4 billion vehicles, are powered by conventional internal combustion engines (ICEs). The consumption of liquid fuel scored about 103.97 million barrels per day in 2022 (Statista, 2021). That is why global transport emissions reached 7.2 Gt of carbon dioxide (CO_2) in 2020. According to one study, air pollution was responsible for approximately 6.7 million deaths worldwide in 2019 (Fuller et al., 2022). Figure 1 presents CO_2 emissions from electricity and heat production by fuel and share by fuel from 2000 to 2021. As shown, coal used for heating and electricity generated 10.5 Gt CO_2 (about 25%) in 2021 (Al-Shetwi, 2022).

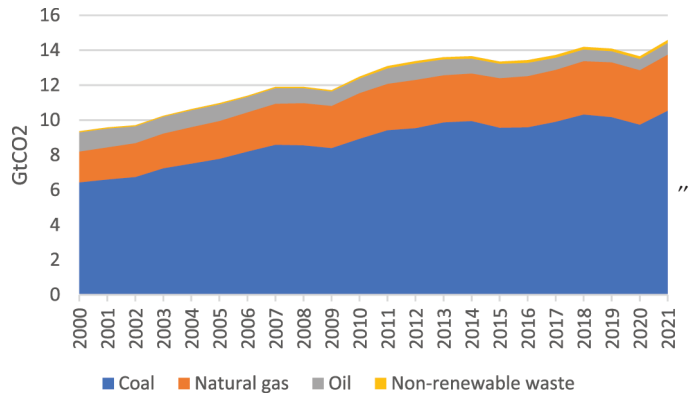
However, the transportation sector contributes significantly to global warming and has potentially disastrous environmental consequences, as shown in Figure 2 (International Energy Agency [IEA], 2020).

Maritime transport, as one of the most important sources of transportation, dominates the international world trade movement, transporting approximately 90% of the world's goods (United Nations Conference on Trade and Development, 2021). Ship movement contributed 2.4% of global CO_2 emissions in 2018, and the percentages of NOx and SOx emissions increased to 15% and 4%–9%, respectively, in 2010 (Ammar & Seddiek, 2020). The International Maritime Organization (IMO) has created new regulations to limit greenhouse gases (GHGs).

SOx emissions are dependent on the percentage of sulfur content in the fuel; this is harmful to human health and causes respiratory and lung diseases. The IMO adopted Regulation 14 of MARPOL Annex VI in 2005 to limit the sulfur content in the fuel that is used on board the ship. In 2020, IMO developed new limits for sulfur known as "IMO 2020," which require reducing the global sulfur ratio from 3.5% to 0.5% (m/m) and 0.1% for ships that sail in emission control areas (Elgohary, Seddiek, et al., 2015). For NOx emissions, the IMO adopted the NOx Technical Code, which came into force on October 10, 2008, to limit NOx emissions (Elgohary, Ammar, et al., 2015). This code applies to marine diesel engines of more than 130 kW; a range of control levels (tiers) apply based on rotational speed (n) and ship construction date. CO_2 emissions are proportional to the fuel consumption rate that contains carbon in its components, and since almost all marine fuels comprise carbon atoms, CO_2 formation is unavoidable. As a

FIGURE 1

CO_2 emissions production by fuel type from 2000 to 2021.

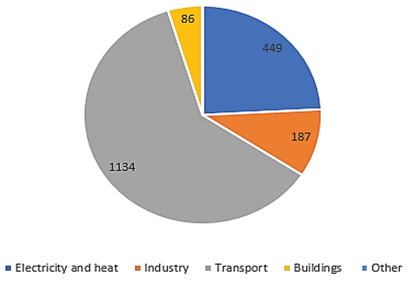


result, the IMO established regulations to limit CO_2 emissions. For new ships, the Energy Efficiency Design Index measure is adopted to promote operating energy-efficient ships by calculating fuel consumption of the propulsion unit in grams of CO_2/ton-mile (Elkafas et al., 2020). For existing ships, the IMO has introduced the ship energy efficiency management plan using the energy efficiency operational indicator to regulate monitoring, reporting, and verification during the operation of ships by measuring the GHG emissions per the transport work.

These restrictions pushed the ship operators to move toward new ways to reduce emissions, namely, using marine gas oil with low sulfur content or additional technology such as scrubbers. However, this increased the capital and operational cost of propulsion unit on board. The innovations used to lower marine emissions are shown in Table 1 (Parker et al., 2015; Andersson & Salazar, 2015). Among these innovations, using alternative renewable fuels is regarded as the most significant method to reduce emissions (Elkafas et al., 2021). There are two main clean and renewable fuel options available: biofuels and electrofuels.

FIGURE 2

Annual change in CO_2 emissions (Mt) by sector in 2020.



The goal of this review article is to investigate, analyze, and provide a concise overview of the most common renewable green fuel sources that are currently being investigated as potential alternative fuels, energy carriers, or storage. We begin by examining biomass as a renewable energy source for producing biofuels via common thermochemical and biochemical processes, and then we demonstrate an electrofuel production strategy as another way of limiting emissions, as well as the key parameters of carbon capture and hydrogen production. Finally, we use methanol and hydrogen as case studies to illustrate the environmental and economic impact of green fuels on air pollution.

Biomass

Biomass is a renewable organic and inorganic solid material produced by natural and technological processes. About 4,500 EJ of solar energy and 120 Gt of carbon from the atmosphere are converted annually into biomass by plants (Rossmeisl et al., 2005). Biomass includes many natural sources such as wood, agriculture crops, biogenic materials, and waste, in addition to animal manure and technological sources such as processing the above natural constituents (Ibrahem, 2012). Biomass has attracted attention lately because of the diminishing fossil fuel reserves and the ever-increasing greenhouse effects caused by fossil fuel use. Figure 3 shows the global increase in total domestic supply of biomass from 42.5 EJ in 2000 to 55.6 EJ in 2018, with solid biomass sources accounting for 85% of total biomass, while liquid biofuels, municipal and industrial waste, and biogas accounted for 7%, 5%, and 3%, respectively (Ben-Iwo et al., 2016).

Biomass is considered one of the earliest sources of energy that can be used to generate electricity, heat, and transportation fuels (Sertolli et al., 2022). According to various reports, our planet stores a massive amount of biomass in various areas ranging from forests to oceans, totaling around 1.8 trillion tons on land and 4 billion tons in sea. In terms of energy, the total biomass production capacity is 33,000 EJ, which is more than 80 times the world's consumption (Tursi, 2019). More than two thirds of renewable energy is generated by biomass, which includes industrial and municipal waste, solid biofuels, and liquid biofuels. From 2000 to 2017, the amount of electricity generated by biomass consumption (bio-power) increased by 3.6 times, and bio-power recorded 596 TWh globally

TABLE 1

CO_2, NOx, and SOx reduction technologies and methods.

Type	Type of operation			
CO_2 emissions	Relate with design	Engine and engine room modifications	Operational measures	New technologies
	Ship weight reduction Redesigned the propeller Ship dimension optimization Bulbous bow optimization	Waste heat recovery Engine adjustments Auxiliary system modification	Hull cleaning Ballast optimization Ship speed reduction Weather forecasting	Carbon capture and storage Alternative fuels Air lubrication system
NOx emissions	Pre-combustion" Engine modification Water injection into the air system Emulation system Using alternative fuels	During combustion: Exhaust gas recirculation system Water addition into cylinder		After treatment: Selective catalytic reduction system
SOx emissions	Using low-sulfur heavy fuel oil (HFO) or marine diesel oil (MDO) Using scrubber system or using sulfur-free alternative fuels			

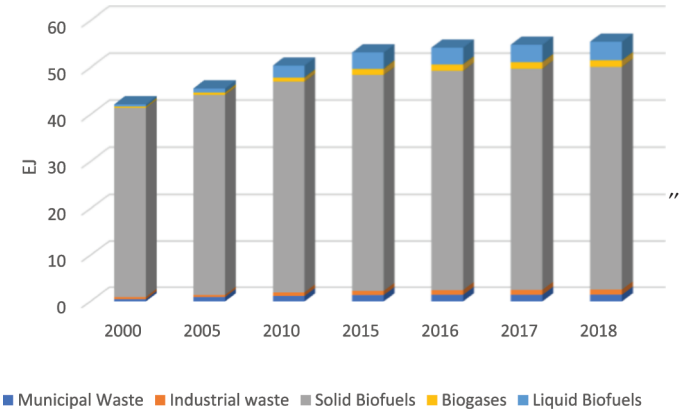
Adapted from Verhelst et al. (2019), Parker et al. (2015), and Andersson & Salazar, (2015).

(bp, 2021). In 2020, total power generation by biomass was about 718 TWh, and it is expected that the value will reach 1,407 TWh in 2030 (EL-Seesy et al., 2022).

Biofuels have properties comparable to traditional fossil fuels and some that are superior, such as a high flash point, lower CO_2 emission, high combustion efficiency, and highcetane number (De Oliveira & Coelho, 2017). However, this process pollutes the environment and is inefficient because the full potential of biomass as an energy source is not utilized. For this, a sustainable way is used to convert biomass into gaseous and liquid fuels with a higher calorific value, such as thermochemical and biochemical processes. This review attempts to assess the current strategies used to convert biomass into biofuels in a comprehensive manner.

FIGURE 3

Global electricity generation from biomass sources.



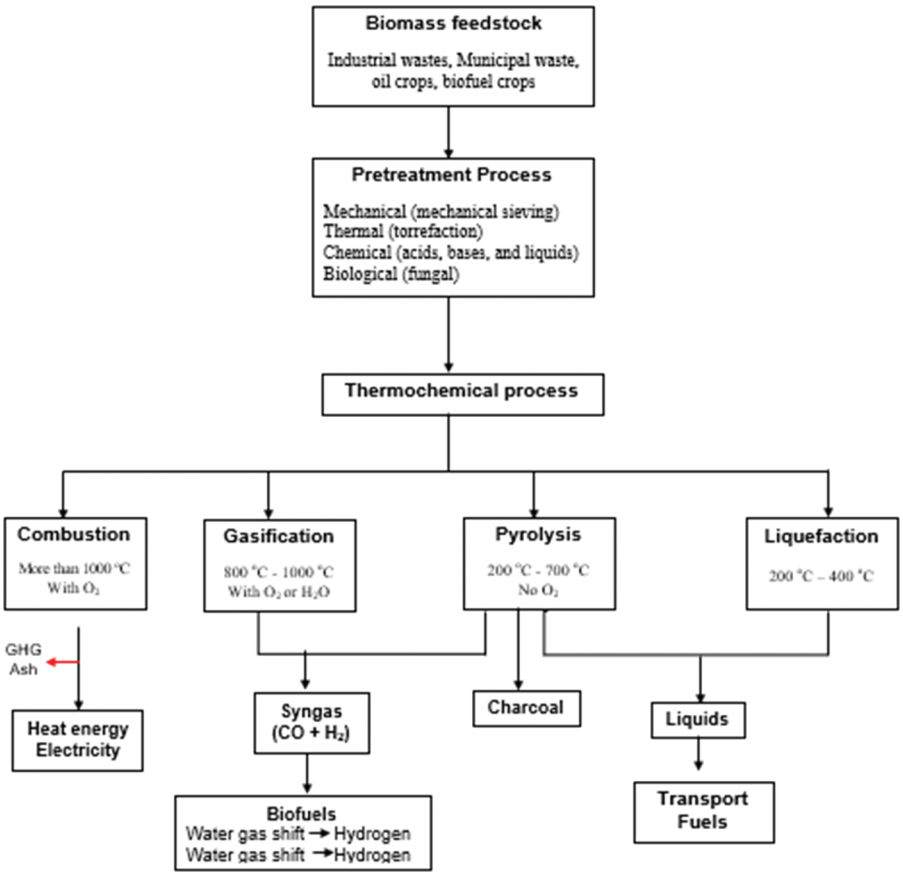
Thermochemical Processes

The thermochemical process is an effective strategy for transforming biomass into biofuels. In this process, biomass thermally decomposes to produce a mixture of H_2 and CO, commonly referred to as "synthesis gas" (syngas). The syngas is then additionally processed to generate liquid fuels such as diesel, methanol, and liquefied hydrogen. As illustrated in Figure 4, the thermochemical process includes combustion, gasification, pyrolysis, and hydrothermal liquefaction for liquid fuel (Ram & Mondal, 2022).

Gasification is a thermal procedure that converts biomass into carbon monoxide, hydrogen, and carbon dioxide, which are then processed to produce biofuels by heating at high temperatures above 727°C (Puig-Arnavat et al., 2010). The gasification process is comparable to combustion, but it has several advantages, such as high thermal efficiency and precise combustion control, in addition to its low CO_2 emissions and the more flexible use of solid by products.

FIGURE 4

The thermochemical process for solid, gas, and liquid productions.



There are two methods for gasification: steam gasification and supercritical water gasification (Boukis & Stoll, 2021; Cao et al., 2020). The biomass gasification process includes pre-treatment, gasification, and gas cleanup steps, in addition to further processing for biofuel production. The quality of the syngas is determined by the biomass feedstock, heat transfer rate, gasifying agent, and reactor type (Ranzi et al., 2016). The gasification agents used in biomass gasification can be classified as air/oxygen, steam, carbon dioxide, or supercritical water gasification (Almena et al., 2022). The gasifier types have a significant effect on the process. Table 2 shows the distinction between them. Another effective way to improve biofuel production from biomass gasification is by using catalysts (Ruhul et al., 2015). As with reactants, catalysts may be homogeneous or heterogeneous (Almena et al., 2022). Two ways to add catalysts into the process are by directly mixing them with raw material or by placing them in a second reactor (Grams & Ruppert, 2017; Nieva et al., 2014).

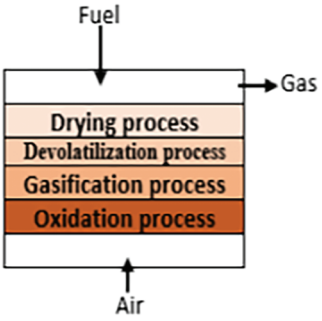
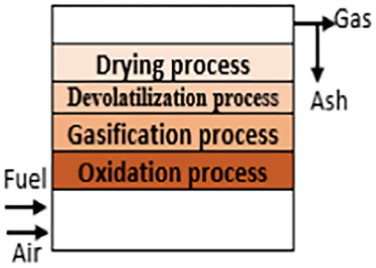
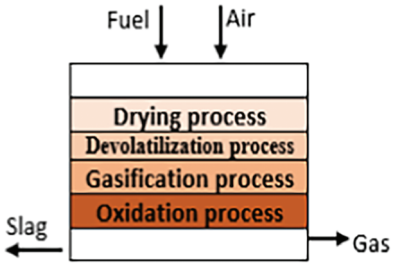
Pyrolysis Process

This is a thermochemical process in the absence of oxygen at 300–800°C to convert biomass into gasses (H_2, CH_4, CO, and CO_2), liquids (tar and oils like acetone, acetic acid, etc.), and solids (char and pure carbon) (Ni et al., 2006). Pyrolysis occurs in three stages: pre-pyrolysis at 120°C–200°C to produce a small amount of CO, H_2O, and CO_2; main pyrolysis to decompose the solid; and char devolatilization. Based on heating rate, particle size, temperature, and solid residence time, biomass pyrolysis can be classified into three types: slow (conventional), fast, and flash pyrolysis (Shen & Yoshikawa, 2013). Slow pyrolysis is not normally suitable for hydrogen production because it is performed at low temperatures (27°C–427°C), has a long residence time, and requires extra energy input. Fast pyrolysis, on the other hand, uses a high temperature to quickly heat biomass without the use of air. Bio-oil production via the fast pyrolysis process has numerous advantages, including low production costs, the ability to store and transport liquid fuel, a neutral CO_2 balance, and high energy density (Jahirul et al., 2012). Flash pyrolysis is regarded as a promising method for improving biofuel production, with the potential to increase production by 75%. However, the process has some production limitations, such as oil corrosivity, low thermal stability, and high viscosity over time (Qureshi et al., 2020; Rasaq et al., 2021). Table 3 shows the difference among types of biomass pyrolysis (Bridgwater, 2018).

Hydrothermal liquefaction is a thermochemical process that converts biomass into liquid fuel by using water at 280°C–370°C and high pressure (10–25 MPa) for an adequate amount of time to crack down the solid biopolymeric structure to liquid components (Toor, 2010). Liquefaction is classified into two types: direct and indirect. Direct liquefaction is a chemical process that breaks down biomass into small molecules in any suitable solvent under high pressure

TABLE 2

Characteristics of the gasifier types.

Types	Fixed bed gasifier	Fluidized bed gasifier	Entrained flow gasifier
Process			
Types	Updraft/downdraft	Bubbling/circulation	Opposed jet/coaxial downflow
Characteristics	Good solid gas contact High resident time and tar content (updraft)	Good solid gas contact Medium resident time Low tar content	Good solid gas contact Seconds resident time Low tar content
Reaction temperature	1090°C	1000°C	1990°C
Exit temperature	650°C	1000°C	1260°C
Advantages	Simple low expensive process Low tar (for downdraft) Small-scale applications	Large-scale applications Generate gas with a higher heating value Easy temperature control	Potential for low tar Generate higher heating value gas
Problems	Feed size restrictions Possible clinkering	Increased particle loading Corrosion and attrition problems	Feed size restrictions
Applications	Small capacities	Medium-size capacities	Large capacities

Adapted from from Ruhul et al. (2015).

(5–20 MPa) and low temperature and then polymerizes these molecules into oily compounds. Pyrolysis, liquefaction, solvolysis, and high-pressure hydrothermal/solvolytic liquefaction producing liquid tars and oils and/or condensable organic vapor are examples of direct liquefaction, while indirect liquefaction is a type of condensing procedure of a gas mixture (CO + H_2) to produce liquid fuels in the use of catalysts. The main parameters that

TABLE 3

The range of important operating parameters for pyrolysis processes.

Biomass pyrolysis	Pyrolysis temperature (°C)	Heating rate (°C/s)	Particle size (mm)	Solid residence time (s)	Production
Slow pyrolysis	300–700	0.1–1	5–50	300–550	Liquid (30%) Char (35%) Gas (35%)
Fast pyrolysis	600–1,000	10–200	<1	0.5–10	Liquid (50%) Char (20%) Gas (30%)
Flash pyrolysis	800–1,000	>1,000	<0.2	<0.5	Liquid (75%) Char (12%) Gas (13%)

Adapted from (Rasaq et al., 2021, and Bridgwater, 2018).

influence the precision of biofuels produced by biomass liquefaction are heating conditions such as temperature and reaction time, biomass type, catalyst, solvent type, and atmosphere (Zhou et al., 2020). Compared to other processes, liquefaction pyrolysis has superior properties. However, there are some limitations to the liquefaction process, such as the need for more complex fuel feeding systems and more expensive reactors.

Biochemical Process

A biochemical system is a chemical process that uses enzymes from bacteria or other microorganisms to degrade biomass into available carbohydrates that can then be converted into liquid fuels, bio-gasses, and various other bio-products. The most common biochemical technologies are anaerobic digestion and fermentation (H. Chen & Wang, 2017). Anaerobic digestion is a biochemical process that occurs without air (anaerobic conditions) for the microbial conversion of biomaterials into alcohols and biofuels (Tursi, 2019). The ability to use any organic material, such as food, industrial and municipal solid wastes, and sewage, is one of the economic benefits of anaerobic digestion. A wide range of fuels can be produced including liquid fuels such as methanol, ethanol, and biodiesel, as well as gaseous fuels such as methane and hydrogen (H. Chen & Wang, 2017). Fermentations are biochemical processes comprising a series of biochemical reactions in which microorganisms, primarily yeasts, convert simple sugars into ethanol and CO_2 under anaerobic conditions (Patinvoh & Taherzadeh, 2019). These techniques are appealing for biofuel production because they use solid waste and provide cheap electricity production while also treating waste. Table 4 shows a comparison of thermochemical and biochemical

TABLE 4

Comparison of the thermochemical and biochemical techniques for biofuel production.

	Combustion process	Gasification process	Pyrolysis process	Fermentation process	Anaerobic digestion
Process	Thermochemical	Thermochemical	Thermochemical	Biochemical	Biochemical
Typical temperature	850°C–1200°C	850°C–1500°C	200°C–750°C	28°C–33°C	35°C
Oxidizing agent	Greater than stoichiometric oxygen	With oxygen or steam	Free of oxygen or steam condition	Free of oxygen or steam condition	Free of oxygen or steam condition
Products	Heat	Heat, fuel gas such as methanol	Bio-oil, fuel gas such as hydrogen	Biodiesel and ethanol	Biogas
Net efficiency	30%–35%	70%–80%	60%–70%	65%–75%	30%–35%
Advantages	- Reduce volume to 90% - Feedstock quality not important	- High efficiency - Reduce emissions - Produce syngas	- Zero waste process - Reduce emissions - Reduce volume to 90%	- Simpler process - Low of capital and operation cost	- Available resources - Low of capital and operation cost
Disadvantages	- High air pollution - Low energy efficiency	- High operation cost - Complex technology	- High investment cost - Extensive gas cleaning	- Environmental - Sensitivities	- Environmental - Sensitivities
Applications	Heating and electricity	Heating, electricity, biochar, and chemical	Heating, electricity, fuel, and chemical	Fuel and chemical	Electricity
The global marketing	–	About 114 operational biomass gasification plants globally, with the value of marketing of US$ 98.2 billion in 2020	Currently approximately 100 pyrolysis plants, with global pyrolysis oil marketing worth US$ 302.1 million in 2020	–	–

Adapted from Molino et al. (2018) and Walling et al. (2019).

techniques for producing biofuels. Although biomass has numerous benefits as a clean energy source for reducing emissions, it is not used commercially because of the challenges associated with its production process management and conversion technologies. Based on the literature research, we can summarize the challenges that face biofuel production as follows (Sansaniwal et al., 2017):

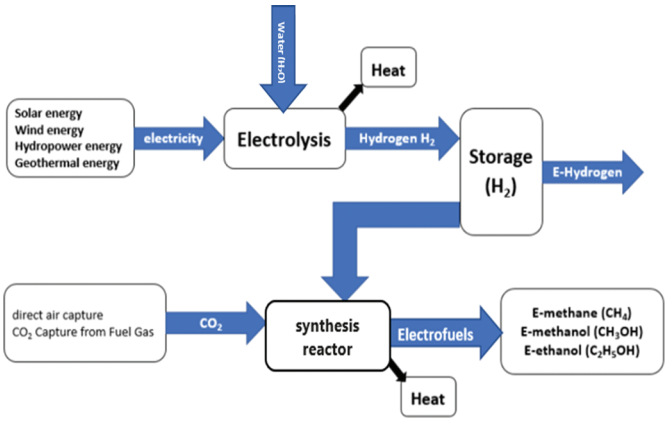
- The moisture content of biomass: It is critical to keep moisture in biomass below 20% because it reduces the fuel consumption potential. This amount of energy is used in the fuel to remove the water. In addition, drying biomass before use raises the overall cost of the operation (Irmak, 2019).
- The biomass properties are very complex; they vary according to conditions such as plant species and their growing location. The ability to measure biomass properties accurately is thus critical when constructing bioprocessing functions.
- The lower biomass density: Overall, biomass density is low, which has a significant impact on transportation and storage costs. Moisture content, size, shape, and surface characteristics are all factors that influence biomass density.
- The moral quandary of food versus fuel: Increasing demand for biomass as a fuel source has a significant impact on starvation, and it is not a suitable option for growing energy crops.

Electrofuels

Electrofuels are another type of renewable fuel. These fuels are based on two parameters: carbon dioxide (CO_2), which is captured from the exhaust of fuel combustion, or directly from the air, in addition to the hydrogen that it generates by using electricity as a primary source of energy. Electrofuels can be converted into a variety of energy carriers including methanol, ethanol, and diesel (Brynolf et al., 2018). Figure 5 depicts the flow diagram for producing electrofuels (Okesola et al., 2018).

FIGURE 5

Process steps in the production of electrofuels.



Carbon Capture (CO_2) From Combustion

For CO_2 capture, three methods can be achieved. Pre-combustion capture includes an integrated gasification combined cycle system that reforms fuels to generate syngas (CO_2 + H2) and then removes CO_2 from the gas before consumption. In post-combustion processes, CO_2 is separated from combustion exhaust gases and then captured using one of many separation methods. In oxy-fuel combustion, pure-oxygen fuels are burned and stored with the exhaust emissions that include a high ratio of CO_2. The main problem for CO_2 capture technology is the cost, because it requires a significant amount of parasitic power with a high electricity cost (Figueroa et al., 2008). Table 5 shows the comparison among various CO_2 capture approaches (Sanz-Pérez et al., 2016).

Direct Air Carbon

Direct air carbon (DAC) technologies are based on extracting CO_2 directly from the air, which can then be used to generate syngas when combined with hydrogen to produce renewable fuels. Two methods are used to capture CO_2 from air directly: a liquid system in which air flows through chemical solutions, such as a hydrostatic solution, to eliminate CO_2, and a solid system in which a solid sorbent filter is used to bind chemically with CO_2. The variation strategies are shown in Table 6. Because of the low concentration of CO_2 in the air (300 times lower than flue gas), a large amount of energy is required to separate it, so the total cost for CO_2 production by DAC is high (Malik, 2019). Another reason for the high cost is the limited CO_2 markets and the fact that there are currently very few companies. Depending on the technology used, costs today range between $100 and $1000/t CO_2. Nowadays, there are 19 DAC

TABLE 5

Comparison among different CO_2 capture approaches.

	Post-combustion capture	Pre-combustion capture	Oxy combustion
Method	Remove CO_2 from the fuel after combustion	Remove CO_2 before it is burned.	Removing CO_2 from fuel with pure oxygen greater than 95%
Concentration	5%–15%	40%	90%
Advantages	Large potential to develop strategy to improve efficiency and reduce cost	Removing CO_2 from syngas increases the amount of CO_2, the driving force for separation, and a wide range of technology for separation.	Removing CO_2 from fuel gas, which is pure oxygen, increases the amount of CO_2 and a wide range of technology available.
Disadvantages	Low concentration of CO_2, low pressure of CO_2 production than sequestration requirements	High cost of equipment, availability	High cost due to O_2 production and additional auxiliary load
Technologies	Amine-based systems: React with CO_2 to form water-soluble compounds.	IGCC: integrated gasification combined cycle to separate CO_2 from syngas	A cryogenic air separation unit It is used to separate O_2 from air and then mixing with recycled fuel gas.
	Carbonate-based systems: React with CO_2 to form bicarbonate such as a K_2CO_3-based system	Physical solvent: Using a physical solvent to absorb CO_2 depends on the solvent type, the partial pressure of CO_2, and the temperature.	
	Aqueous ammonia: Ammonium carbonates react with syngas to produce bicarbonate.		
	Membranes: such as using a bundle of membrane tubes, an inorganic membrane	Membrane: such as a polybenzimidazole	Oxygen transport membrane is another method that allows pure oxygen to transport into the furnace.
	CO_2 capture solid sorbents: include a wide range of porous, solid phase materials, such as mesoporous silicas and zeolites	Sorbent materials: such as e-lithium silicate based (Li_4SiO_4) sorbent material	
	Enzyme-based system: It is a biological system based on natural reactions of CO_2 in living organisms to improve CO_2 capture.	Loop combustion and gasification: using solid oxygen for combustion instead of air to produce concentrated CO_2 without the need for any separation unit	Another method to separate oxygen before combustion is ceramic autothermal recovery, which absorbs O_2 from the air in a perovskite fixed bed.
	Ionic liquids have the potential to CO_2 capture better, which can lower total electrical.		

Adapted from (Bartels et al., 2010; Gutiérrez-Sánchez et al., 2022; T. Li et al., 2016; and Sanz-Pérez et al., 2016).

systems globally producing more than 0.01 Mt CO_2/year, and this capacity is expected to expand to 85 and 980 Mt CO_2/year in 2030 and 2050, respectively (IEA, 2022a). Figure 6 depicts the carbon capture, utilization, and storage facilities in use by application from 1980 to 2021 (IEA, 2021). The results are provided in terms of million tons of CO_2.

Water Splitting (Electrolysis)

Electrolysis is the most common process for producing hydrogen from water splitting; it uses two separated electrodes: a cathode and an anode (Brynolf et al., 2018). It is an electrochemical process in which decomposed water is converted by passing a current through it into positive charge H^+, which moves to the negative cathode, and negative charge O^-, which moves toward the positive anode, and then H_2 is formed. This can be accomplished with three types of electrolyzers: alkaline, proton exchange membrane (PEM), and solid oxide, as shown in Table 7 (Nikolaidis & Poullikkas, 2017). Alkaline electrolysis is popular because it is low in capital costs and

TABLE 6

The difference between liquid and solid systems for direct air capture based on using natural gas as a source for heat and electricity energy and for plant produce 1 Mt CO_2/year.

	Liquid systems	Solid DAC technology
Energy to storage, GJ/t CO_2	Heat energy = 5.25 Electricity energy = 1.32	Heat energy = 7.20 Electricity energy = 2.82
Energy to use, GJ/t CO_2	Heat energy = 5.25 Electricity energy = 0.84	Heat energy = 7.20 Electricity energy = 2.34
CO_2 generated, Mt CO_2/year	Heat source = 0.47–0.66 Electricity source = 0.11–0.23	Heat source = 0.22–0.3 Electricity source = 0.07–0.14
Capture cost, $/t CO_2	147–264	88–228

Adapted from IEA, (2022a) and Malik (2019).

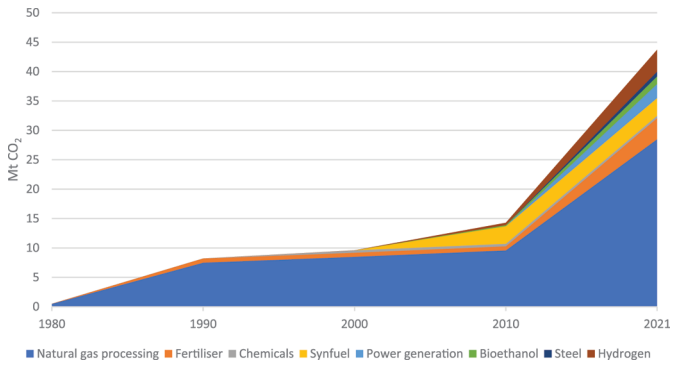
has the availability of large plant sizes in addition to a long lifetime, so it has a larger commercial outreach (Ebaid et al., 2015). However, there are some disadvantages, such as low density, high maintenance costs, and storing enough hydrogen for the process. An additional compressor is required because alkaline reduces efficiency during storage. PEM electrolyzers use a thin polymeric membrane as an electrolyte instead of a liquid electrolyte in alkaline electrolysis. To improve the efficiency of membranes, noble metals such as platinum and iridium are used, but they are highly expensive. PEM electrolyzers can operate at pressures of more than 100 bars, and no corrosive substances are involved (David et al., 2019). A solid oxide electrolyzer is the third method to split water for hydrogen production. It takes approximately 50 kWh of electrical energy to electrolyze 9 L of water to produce 1 kg of hydrogen. According to detailed studies conducted by the National Renewable Energy Laboratory, the electrolyzer infrastructure costs about $0.80 per kg of H_2 produced.

Methanol Fuel

Methanol is the first alternative fuel used as a case study in this article.

FIGURE 6

CCUS facilities in operation by application from 1980 to 2021.



Although methanol combustion reduces the NOx and SOx emissions by 45% and 8%, respectively, the environmental studies show that creating methanol via a conventional method produces high GHG emissions compared with using marine diesel oil and heavy fuel oil (IMO, 2016). Studies and research have moved toward using renewable methods such as biomass resources to produce methanol. The results confirmed the potential of biomethanol to save significant emissions based on the source of electricity used, so the source of electricity is regarded as a crucial factor for the total bio-methanol GHG emissions (Brynolf et al., 2014). Around 200,000 tons of bio-methanol were produced in 2013. Dowaki and Genchi (2009) estimate that producing biomethanol from biomass, such as Japanese wood, could reduce CO_2 emissions by 24%–40%. In addition, some 2010 studies in Germany showed that biomethanol production by forest waste and short rotation coppice reduced emissions by 24%–33% (Broeren et al., 2013).

Currently, green methanol is considered to meet IMO regulations for several emissions, which could reduce SOx to less than 0.1, complying with Sulfur Emission Control Area (SECA), reducing 80% of NOx emissions and 95% of particulate matter, in addition to reducing CO_2 by 15%. Biomethanol production costs depend on several key factors, such as biomass feedstock types and prices, and the fuel prices that are used to generate electricity (Clausen et al., 2010). Using renewable electricity will increase capital costs from 23% to 65% of the total production cost, depending on the local conditions of the country (Broeren et al., 2013).

TABLE 7

Characteristics of alkaline, PEM, and solid oxide water electrolysis.

Types	Alkaline electrolysis (AEL)	PEM electrolysis	Solid oxide electrolysis (SOEL)
Chemical formula	Types / Anode / Electrolyte / Cathode AEL: $2OH^- \rightarrow H_2O + \frac{1}{2}O_2 + 2e^-$; OH^- ← ; $2H_2O + 2e^- \rightarrow 2OH^- + H_2$ PEM: $H_2O \rightarrow \frac{1}{2}O_2 + 2e^- + 2H^+$; H^+ → ; $2H^+ + 2e^- \rightarrow H_2$ SOEL: $O^{-2} \rightarrow \frac{1}{2}O_2 + 2e^-$; O^{-2} ← ; $H_2O + 2e^- \rightarrow O^{2-} + H_2$		
Temperature	60°C–80°C	50°C–80°C	600°C–900°C
Advantages	High potential power up to 100 MW Low raw material cost Established technology	High-purity hydrogen Excellent dynamic properties Gradient of high load	High efficiency up to 80% Suitable for co-electrolysis
Disadvantages	Vulnerability to impurities in manufacturing gases Low dynamic	High investment costs Require precious metals Acidic corrosive	High investment costs High cold start time

Adapted from Nikolaidis & Poullikkas, (2017).

According to International Renewable Energy Agency (IRENA) estimations, the cost of methanol production from natural gas ranges from EUR 75/t to 250/t, and from coal, it is EUR 150/t to 300/t. The total global weight cost is estimated to be EUR 160/t. The capital cost of biomethanol production is 3.4 times higher than the capital cost of plants based on natural gas (Bromberg & Cheng, 2010). As mentioned above, the production of e-methanol is based on two parameters: CO and H_2. Converting one mole of CO_2 to methanol requires about 228 KJ of energy input in addition to six electrons reducing C^{4+} to C^{2-}, and this is expressed in the equation (Ravikumar et al., 2020):

$$CO_2 + 3H_2 \rightarrow CH_3OH + H_2O$$

E-methanol, also known as renewable methanol, has the potential to reduce carbon emissions in marine transportation by more than 90% while also providing a sustainable alternative to fossil fuels. Several studies have been conducted to improve the chemical method for e-methanol production (Van-Dal & Bouallou, 2013). In 2022, hybrid e-methanol production with biomass gasification was assessed, and the results show the environmental and economic benefits of using CO_2 exhaust from gasification for e-methanol production, with an overall efficiency of 85.8%, in addition to cost savings when compared to pure e-methanol production (K. Chen et al., 2015). As shown in Table 8, several pilot projects are evaluating various options for the adoption of e-methanol as alternative fuels to enhance GHG reduction (IRENA & Methanol Institute, 2021).

Hydrogen Fuel

The second case study is hydrogen fuel. Hydrogen is a non-producer of emissions from the vessel operating phase due to the absence of carbon, in addition to possessing the highest energy content of 120 MJ/kg (El Gohary et al., 2014). It can be used as a fuel for ICEs or in a fuel cell to produce electricity (IEA, 2022b). Hydrogen is also used domestically because it can be used as a clean transportation fuel to generate electricity for the grid and to offset the carbon footprint of industrial processes. Table 9 depicts the various methods for producing hydrogen. Hydrogen is primarily produced from natural gas and coal; thus, the process emits approximately 25 million metric tons of carbon dioxide per year.

Away from fossil fuel, hydrogen can be produced from various sources, such as renewable electricity (ehydrogen), biomass (biohydrogen), and nuclear power. Biohydrogen is a green fuel with low emissions. Using carbon capture and storage with biohydrogen production could generate

TABLE 8

Renewable methanol projects.

Company	Year of start	Feedstock	Capacity (ton/year)	Products	Environmental effect
CRI, Finnfjord, and Statkraft	2024	CO_2 from a ferrosilicon plant and H_2 from water electrolysis (hydropower)	100,000	E-methanol	Reusing 300,000 tons of CO_2 per year, which is equivalent to permanently removing over 100,000 fossil-fueled cars
Reintegrate	2023	CO_2 from a biogas plant and H_2 from water electrolysis	10,000	E-methanol	To produce CO_2-neutral jet fuel
Hy2gen	2025	CO_2 from biogas and H_2 from water electrolysis	61,000	E-methanol	Eliminate approximately 8.96–12.60 tons of CO_2 per ton of hydrogen
Air company	2020	Biogenic CO_2 and H_2 from water electrolysis	1,000	E-methanol	The goal is to avoid the equivalent of 10.8% of global CO_2 emissions or more than 4.6 billion tons of CO_2 per year.
Wacker Chemie and Lind	Planning	CO_2 and H_2 from water electrolysis	15,000	E-methanol	Reducing CO_2 emissions by approximately 80% when compared to the current production methods
Vicat and Hynamics	Planning	CO_2 from a cement plant and H_2 from water electrolysis	200,000	E-methanol	The project aims to capture 40% of the CO_2 emitted by Vicat, a Montalieu-Vercieu cement plant in eastern France.
Liquid wind	2024	Upcycle CO_2 and H_2 from water electrolysis	50,000	E-methanol	By replacing fossil fuels, it is possible to reduce carbon emissions by 100,000 t.
Perstorp	2025	CO_2 and H_2 from water electrolysis	200,000	E-methanol	Attempt to reduce greenhouse gas emissions by 500,000 tons per year
CRI and Jiangsu Sailboat Petrochemical	2023	CO_2 and H_2 from water electrolysis	100,000	E-methanol	To produce 100,000 tons of methanol per year, the plant will recycle about 150,000 tons of CO_2 and 20,000 metric tons of H_2 from other onsite processes.
Iberdrola and Foresa	Planning	CO_2 and H_2 from water electrolysis	100,000	E-methanol	The aim is to increase green methanol production from 10,000 tons per year to 100,000 tons per year. This prevents 470,000 tons of CO_2 emissions per year.

Adapted from IRENA & Methanol Institute, (2021).

TABLE 9

Comparison of different hydrogen production processes.

Process	Efficiency %	CO_2/H_2 (kg/kg)	Cost for H_2 ($/kg)	
Conventional method				
Steam reforming	74–85	10–11	2.27 without CC 2.08 with CC	
Partial oxidation method	60–75	9–11	1.34–1.63	
Autothermal method	60–75	9–11	$13.48/GJ or 1.48	
Hydrocarbon pyrolysis	–	9–11	1.59–1.70 for methane	
Renewable resources				
Biomass pyrolysis	35–50	1–2	1.25–2.20, depending on the facility size and biomass type	
Biomass gasification	65	1–2	1.77–2.05	
Water splitting (electrolysis)	40–60	–	Solar PV	5.78–23.27
			Solar thermal	5.10–10.49
			Wind	5.89–6.03
			Nuclear	4.15–7.00
Water splitting (thermochemical)	20–45	–	Solar	7.98–8.40
			Nuclear	2.17–2.63
Water splitting (photolysis)	7.8	–	10.36	

Adapted from Acar and Dincer (2018), Kayfeci & Keçebaş (2019), and Steinfeld (2005).

net-negative CO_2 emissions (Parkinson et al., 2019). Fast pyrolysis produces approximately 30.7–66.5 g of hydrogen per kilogram of biomass, whereas steam gasification and supercritical water gasification produce approximately 40–70 g and 30–50 g, respectively (Cao et al., 2020). Prasertcharoensuk et al. (2019) investigated the feedstock parameters for the gasification process that affects hydrogen production and found that with pyrolysis temperatures greater than 800°C and particle sizes of 0.5–1 cm^3, the H_2 content in syngas increased by up to 67 mol% (Prasertcharoensuk et al., 2019). H. Li et al. (2015) also showed that increasing temperature has a significant effect on gas yield and hydrogen concentration (H. Li et al., 2015). Chang et al. (2011) studied biohydrogen production by biomass gasification and used a-cellulose and other agricultural waste as a biomass feedstock at temperatures of 600°C–1000°C. Despite the advantages of biohydrogen production over fossil fuels, its development is hampered by critical limitations and obstacles in terms of current financial viability and technological readiness (Osman et al., 2020).

Fuel Cell Technology

A fuel cell is a type of electrochemical energy transformation device that switches potential chemical energy into electrical energy while also producing heat as a byproduct (Sazali et al., 2020). The use of fuel cells dates to 1839. Fuel cell systems provide environmentally friendly, efficient, and dependable power. Fuel cells, unlike batteries, are not required to be recharged on a regular basis but instead continue to produce electricity if a fuel source is available (Bilgili et al., 2015). The basic function of a fuel cell is to oxidize hydrogen through an anode, splitting it into electrons and protons. Electricity is produced by electrons passing through a circuit, while protons move toward the cathode to react with oxygen and electrons to produce water. There are various types of fuel cells, all of which differ in terms of catalytic electrodes, fuel used, temperature, and pressure operation. Globally, over 800 MW of large stationary fuel cell systems with power ratings greater than 200 kW have been installed for distributed generation and combined heat and energy applications (U.S. Department of Energy, 2011). Many research studies have been conducted to investigate the various types of fuel cells and their characteristics, which are illustrated in Table 10 (Mekhilef et al., 2012). Fuel cell technology has proven to be effective in the aerospace, automotive, and power plant sectors, as well as naval vessels. For marine applications, both the molten carbonate fuel cell and the PEM fuel cell (PEMFC) are regarded as the most promising options. They reduce NOx, CO, and CO_2 emissions, as well as vibration and noise, with a thermal efficiency of more than 40%.

Fuel cells can coexist with renewable fuel energy and modern energy sources such as methanol and hydrogen. There are two ways of using methanol as a fuel in fuel cells: direct methanol fuel cells (DMFCs) and

TABLE 10

The different types of fuel cells.

	Alkaline fuel cell (AFC)	PEMFC	Direct methanol fuel cell (DMFC)	Phosphoric acid fuel cell (PAFC)	Molten carbonate fuel cell (MCFC)	Solid oxide fuel cell (SOFC)
Electrolyte	Potassium hydroxide	PEM	PEM	Phosphoric acid	Molten carbonates	ceramic
Reactions	types / Anode electrode / Electrolyte / Cathode electrode AFC: $2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$	OH^- ←	$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$			
	PEMFC: $H_2 \rightarrow 2H^+ + 2e^-$	H^+ →	$\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$			
	DMFC: $CH_3OH + H_2O \rightarrow CO_2 + 6H^+ + 6e^-$	H^+ →	$\frac{3}{2}O_2 + 6H^+ + 6e^- \rightarrow 3H_2O$			
	PAFC: $2H_2 \rightarrow 4H^+ + 4e^-$	H^+ →	$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$			
	MCFC: $CO_3^{-2} + H_2 \rightarrow CO_2 + H_2O + 2e^-$ $CO_3^{-2} + CO \rightarrow 2CO_2 + 2e^-$	CO_3^{-2} ←	$\frac{1}{2}O_2 + CO_2 + 2e^- \rightarrow CO_3^{-2}$			
	SOFC: $O^{-2} + H_2 \rightarrow H_2O + 2e^-$ $CO + O^{-2} \rightarrow CO_2 + 2e^-$	O^{-2} ←	$\frac{1}{2}O_2 + 2e^- \rightarrow O^{-2}$			
	CO: carbon monoxide CO_2: carbon dioxide CO_3^{-2}: carbonate ion		e^-: electron H^+: hydrogen ion H_2: hydrogen		H_2O: water O_2: Oxygen OH^-: hydroxyl ion	
Fuels	H_2	H_2	CH_3OH	H_2	H_2	H_2
Temperature	60°C–90°C	80°C	70°C–130°C	200°C	650°C	1,000°C
Efficiency	45%–60%	40%–60%	40%	35%–40%	45%–60%	50%–65%
Power (kW)	Up to 50	Up to 250	Up to 100	Up to 1,000		Up to 2,000
Advantages	High efficiency	Operate at low temperature	Operate at low temperature	Tolerant to impure H_2	High efficiency Available for many fuels	Available for many fuels High efficiency
Problems	Carbon dioxide sensitive	Expensive catalysts for H_2 production	Slow reaction for moving fuels from anode to cathode	Expensive catalysts for H_2 production	High corrosive catalysts	High operating Temperature
Applications	Space application	Transportation	Vehicle and small portable	Power plants	Power plants	Power plants

Adapted from Mekhilef et al. (2012).

high-temperature PEMFC (HT-PEMFC). DMFCs use liquid methanol directly as a fuel to produce electricity; their advantage is the potential of methanol to be transported and stored easily (Araya et al., 2020). However, there are disadvantages to DMFCs, such as a drop in efficiency of below 30% due to methanol crossover through the membrane. HT-PEMFC is another way to use methanol as a fuel. It operates at temperatures between 160°C and 180°C (Dalena et al., 2018). Hydrogen can also be used in a fuel cell to generate electricity, which applies to vehicles and distributed production. It is expected that hydrogen fuel cells will be a significant factor in the future for low CO_2 due to their ability to harness renewable sources such as solar, wind, wave, and tidal energy for non-toxic fuel source production, and this improves sustainable energy. Since 1980, fuel cells have been used in marine applications, with several projects completed, as shown in Table 11 (Xing et al., 2021). Maritime fuel cells can be used as a primary propulsion force for low power demands, as well as a source of electricity, and as an auxiliary power source for hybrid propulsion (van Biert et al., 2016).

TABLE 11

Marine applications using fuel cells.

Vessel/project	Year	Fuel cell	Fuel	Capacity (kW)	Application
Class 212	1980	PEMFC	Hydrogen	120	Submarine
SSFC	1997	MCFC/PEMFC	Diesel	500	Naval vessel
FellowSHIP	2003	MCFC	(LNG)	320	Offshore supply
Ballard Power Systems	2003	PEMFC	Hydrogen	4.8	Yacht
MCWAP	2005	MCFC	Diesel	150/500	RoPax, RoRo
EU FC program	2008	PEMFC	Hydrogen	100	Small ferry power
Pa-X-ell MS Mariella	2009	HT-PEMFC	Methanol	2 × 30	Cruise ship
Nemo H2	2009	PEMFC	Hydrogen	70	Passenger ships
SHIBZ	2009	SOFC	Methanol	27	Multipurpose
FCSHIP	2010	PEMFC	Hydrogen	400	Small ferry power
METHANPU	2010	SOFC	Methanol	20	RORO/passenger
SF-BREEZE	–	PEMFC	Hydrogen	120	150 passengers
RiverCell	–	HT-PEMFC	Methanol	250	River cruise vessels
MARANDA	2017	PEMFC	Green hydrogen	165	Research vessel Aranda

Adapted from (Elgohary et al., 2014; and Xing et al., 2021).

Conclusions

Because of their low carbon footprint, biofuels and electrofuels are viewed as promising candidates for fossil fuel replacement in a zero-carbon future. This review article provides a brief critical discussion of two alternative green fuels, namely, biofuels and electrofuels, to encourage comparison and achieve the best results, as well as to provide impetus for further research into how to improve the efficiency of each. Finally, the impact of biofuel and electrofuel on maritime applications has been discussed to aid industry and research. The article focused on methanol and hydrogen as a case study. The current study yielded the following conclusions.

- Data show that biofuel production is steadily increasing because it provides greater energy security, lower greenhouse gasses and particulate emissions, improved vehicle performance, and lower demand for petroleum. Combustion, gasification, pyrolysis, and liquefaction are examples of thermochemical processing for biofuel production. Biochemical processing, on the other hand, involves fermentation and anaerobic digestion.
- Temperature, heating rate, reactor size, reactor type, residence time, and catalysts play a significant role in the amount and efficiency of biofuel production, as well as the amount of coke formed during the process.
- Biomass gasification is regarded as the most efficient method for producing biohydrogen, with a 65% efficiency, a cost of $1.77–$2.05, and an emission of 1–2 kg of CO_2 per kilogram of H_2. Despite the environmental and economic benefits of biomass, high moisture rates and low density are significant challenges that limit biofuel production rates.
- The article reviewed the advancements in the field of electrofuels and their innovations. The term "electrofuel" refers to gas or liquid fuels that use CO_2 as the primary carbon source, as well as hydrogen derived from water splitting using a zero-carbon electricity source. According to the technological and financial aspects of electrofuel production, the cost of electricity is the most significant contributor to the final cost of these fuels.
- The study examined the operating and chemical reactions of various fuel cell types. The results show that PEMFCs are better suited to commercial and residential use due to their low operating temperature (50°C–100°C) and quick startup.

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